

The effect of spatial variation in surface relaxivity on nuclear magnetic resonance relaxation rates

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ABSTRACT

Nuclear magnetic resonance (NMR) relaxation measurements are sensitive to the physiochemical environment of water in saturated porous media and can provide information about the properties of geologic material. Interpretation of NMR data typically relies on three assumptions: that pores within the geologic material are effectively isolated such that the diffusion of a proton between pores is limited (i.e., there is weak coupling); that relaxation occurs in the fast-diffusion regime; and that surface relaxivity ρ_2 is uniform throughout the measured volume. We investigated the effect of spatial variation in ρ_2 on the NMR relaxation measurement and evaluated two equations relating ρ_2 to the NMR relaxation rate for samples containing two types of surfaces, each with a different surface

relaxivity. One equation was valid when there is weak diffusional coupling between pores, the other is valid when there is strong diffusional coupling. We prepared a suite of samples composed of quartz sand and an iron-coated quartz sand. NMR relaxation occurred in two distinct regions: the weak- and strong-coupling regions. In the weak-coupling region, the equation did not accurately represent the relationship between the two ρ_2 values and the NMR relaxation rate, suggesting that further research is required to understand the effect of spatially variable ρ_2 in this relaxation region. In the strong-coupling region, the equation accurately represented the relationship between the two values of ρ_2 and the NMR relaxation rate. The results from these laboratory experiments represented a first step towards accounting for spatial variability in ρ_2 in the interpretation of NMR data.

INTRODUCTION

The proton nuclear magnetic resonance (NMR) relaxation measurement is used in earth sciences in the exploration for, and evaluation of, petroleum and groundwater resources. NMR borehole logging instruments have been used since the 1960s to detect water and hydrocarbons and estimate petroleum reservoir properties such as porosity, pore size, and permeability (e.g., Allen et al., 2000). Recently, a new borehole-logging tool has been developed that is designed for use in near-surface sediments to detect water and to estimate water-filled porosity, pore-size distributions, and hydraulic conductivity in groundwater aquifers (Maliva et al., 2009; Walsh et al., 2010). A related field method, surface NMR (SNMR), is used for the same purposes to characterize groundwater aquifers in the top ~150 m of the subsurface using a wire loop laid out on the surface (e.g., Legchenko et al., 2004; Dlugosch et al., 2011). Laboratory studies are used to improve our interpretation

of NMR data and explore new uses for NMR. Recent studies focused on near-surface applications, for example, have highlighted the impact that changing redox conditions in the near-surface environment can have on measured NMR parameters (Bryar and Knight, 2002; Keating and Knight, 2007, 2008, 2010; Keating et al., 2008) and the importance of accounting for the higher degree of pore connectivity often found in unconsolidated near-surface materials (Grunewald and Knight, 2009). **Improving the interpretation of NMR measurements in water-saturated near-surface materials is the application of interest in our research.**

The NMR relaxation experiment, for a water-saturated porous material, consists of observing the response of hydrogen nuclei (protons) in pore water, placed in a static magnetic field, to a perturbation by a secondary magnetic force and measuring a voltage induced by the precessional motion of the protons during their return or relaxation to equilibrium. In the interpretation of NMR data in geophysics, the rate of this relaxation is typically assumed to be

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proportional to the surface-area-to-volume ratio of the pore space (e.g., [Cohen and Mendelson, 1982](#)). The constant of proportionality is the surface relaxivity, which quantifies the ability of the grain surface to enhance relaxation. This relationship between relaxation rate and surface-area-to-volume ratio is the basis for using NMR data to obtain information about the geometry of the pore space and estimate the permeability. The validity of this relationship, however, relies on three critical assumptions. The first assumption is that pores are effectively isolated such that the diffusion of a proton between pores in the time-scale of the NMR experiment is limited; this is referred to as weak diffusional coupling. The situation in which a proton can diffuse between many pores is referred to as strong diffusional coupling. The second assumption is that relaxation occurs in the fast-diffusion regime in which diffusion of protons to the pore surface, where relaxation occurs, is not the rate-limiting process. The third assumption is that the surface relaxivity can be treated as uniform across the entire surface of the pore space in the measured sample. In this study, we focus on this third assumption and consider NMR relaxation for cases in which surface relaxivity is spatially variable. In geologic material, we would expect the surface relaxivity to be nonuniform when there is more than one mineral phase present within a single pore or on the surface of a single grain. Multiple minerals on a single grain have been observed in samples where microbes mediate mineral transformations during the remediation of contaminants (e.g., [Fredrickson et al., 2004](#); [Williams et al., 2009](#)).

The theoretical model of relaxation at the surface of a pore relates the surface relaxivity to the spatial density of paramagnetic species, e.g., unpaired electrons in iron(III), on the solid surface, the time for a proton to move along the surface, and the residence time of a proton at the solid surface ([Gillis and Koenig, 1987](#); [Godefroy et al., 2001](#)). Two recent studies by [Ryu \(2009\)](#) and [Ryu and Johnson \(2009\)](#) rigorously explore the effects of nonuniform surface relaxivity on the NMR relaxation rate using perturbation theory, exact theoretical results and numerical simulations on random sphere packs. The results of these works show that the NMR response of a sample with spatially variable surface relaxivity will depend on the relative magnitude and spatial distribution of the surface relaxivity. In particular, the results from the numerical simulations on random sphere packs show that when the surface of each sphere within the packing is randomly assigned one of two surface relaxivity values, differing by a factor of 11.5, the overall NMR response of the system is very different from the case of uniform surface relaxivity. [Ryu and Johnson \(2009\)](#) conclude that NMR measurements collected on such samples are likely to be misinterpreted.

In this work, we extended upon these previous studies and conducted laboratory experiments that explore the NMR response of water-saturated granular materials containing two types of grains, each with a different surface relaxivity value. We define relaxivity as being uniform if there is one type of grain, so we assume that the surfaces of all the grains have the same mineral composition. We define relaxivity as being nonuniform or spatially varying if there are two types of grains, so we assume that the surfaces are composed of two different minerals. In this study, we created samples with nonuniform surface relaxivity by mixing quartz sand and quartz sand coated with an iron-mineral, referred to as iron-coated quartz sand. We first present two equations representing the relationship between surface relaxivity and NMR relaxation rates for samples with spatially variable surface relaxivity. One represents

the relationship when there is weak diffusional coupling between pores; the other represents the relationship when there is strong diffusional coupling. Using the results from the laboratory experiments, we evaluated the validity of the two equations presented. Additionally, as both equations assume fast diffusion, the assumption of fast diffusion was tested. The results presented in this study comprise the first data set to systematically evaluate equations linking surface relaxivity to NMR relaxation rates in samples with spatially variable surface relaxivity. Evaluating the effect of spatially variable surface relaxivity on the NMR response of water-saturated unconsolidated material is critical if NMR data are to be used to accurately determine pore-size distributions or permeability.

BACKGROUND AND THEORY

The NMR relaxation phenomenon is observed in nuclei with an odd number of protons because these nuclei possess a nuclear spin angular momentum. The hydrogen nucleus, with a single proton, is of particular interest because of its presence in water. In a static magnetic field, the nuclear spins of the protons will align with the field, resulting in a net magnetization proportional to the number of protons in the sample. In the NMR experiment, an oscillating magnetic field, tuned to detect hydrogen, is applied to the sample for a short time causing the nuclear spins to diverge from, and then relax back to, their equilibrium positions. This results in a measurable change in the magnetization over time t .

In this study, we focused on the time-dependent behavior of the transverse magnetization, $M_{xy}(t)$. The behavior of $M_{xy}(t)$ in a single pore, $M_{xy}(\mathbf{x}, t)$, where $\mathbf{x}$ is the location in the pore, is described by the modified Bloch equations ([Torrey, 1956](#)),

$$\frac{\partial M_{xy}(\mathbf{x}, t)}{\partial t} = D \nabla^2 M_{xy}(\mathbf{x}, t) - \frac{M_{xy}(\mathbf{x}, t)}{T_{2B}} - i\gamma \Delta B_z M_{xy}(\mathbf{x}, t), \quad (1a)$$

with the initial condition,

$$M_{xy}(\mathbf{x}, 0) = \text{constant}, \quad (1b)$$

and the boundary condition evaluated at the pore-grain interface,

$$D \nabla (M_{xy} \cdot \hat{\mathbf{n}}) - \rho_2 M_{xy} = 0. \quad (1c)$$

In equation 1, D is the self-diffusion coefficient ($2.5 \times 10^5 \text{ cm}^2/\text{s}$ for water at 30°C), T_{2B} is the bulk fluid relaxation time, γ is the gyromagnetic ratio ($\gamma = 0.267 \text{ rad}/[\text{nT} \cdot \text{s}]$ for protons in water), ΔB_z , which is typically assumed to equal zero, accounts for inhomogeneities in the magnetic field and is the z component of the difference between the static field and the actual magnetic field, $\hat{\mathbf{n}}$ is the vector normal to the surface of the grain, and ρ_2 is the surface relaxivity; the inverse of the bulk fluid relaxation time is the bulk fluid relaxation rate, T_{2B}^{-1} . The measured bulk NMR signal is a function of time, $M_{xy}(t)$, and can be obtained from the volume integral of $M_{xy}(\mathbf{x}, t)$.

[Brownstein and Tarr \(1979\)](#) show that the modified Bloch equations are an eigenvalue problem and define three relaxation regimes for the solution for water relaxing in a single pore. The boundary of each relaxation regime is determined by the control parameter, $\kappa \equiv \rho_2 a / D$, where a is the mean distance that a proton must travel to reach a paramagnetic site; for the commonly assumed case of a

spherical pore a is the pore radius divided by three, equivalent to $V_{\text{pore}}/S_{\text{pore}}$, where V_{pore} and S_{pore} are the volume and surface area of the pore, respectively. The first regime, the slow-diffusion regime, occurs when a diffusing proton does not sample the entire pore space before relaxing at the pore surface, the boundary of the first regime is determined by $\kappa \gg 10$. The second regime, the intermediate-diffusion regime, occurs when $1 < \kappa < 10$. The last regime, the fast-diffusion regime, occurs when a diffusing proton can travel to and relax at the pore surface within the time-scale of the NMR measurement; the boundary of the last regime is determined by $\kappa \ll 1$. It has been suggested that a sufficient condition for fast diffusion is $\kappa < 0.1$ (Ryu and Johnson, 2009).

NMR relaxation in the slow and intermediate diffusion regimes

In the slow- and intermediate-diffusion regimes, the solution for water in a single pore is a sum of exponential decays,

$$M_{xy}(t) = M(0) \sum_{k=0}^{\infty} f_k \exp(-t/T_{2Sk}), \quad (2)$$

where $M(0)$ is the initial magnetization and is proportional to the total number of protons in the pore, f_k is the relative amplitude for each component of the measured signal decaying with time constant T_{2Sk} . A factor of $\exp(-t/T_{2B})$ has been divided out of $M_{xy}(t)$ in equation 2, thus removing a term representing the relaxation of the bulk fluid. The sum is arranged such that $T_{2S0}^{-1} < T_{2S1}^{-1} < T_{2S2}^{-1} < \dots$ and the k th term in the summation is referred to as the k th relaxation mode. In the slow-diffusion regime, most of the magnetization is in the slowest mode (i.e., most of the magnetization relaxes with a rate as of T_{2S0}^{-1}) and between 10% and 40% of the relaxation occurs in higher modes, depending on the characteristic shape of the pore. In the intermediate diffusion regime, most of the magnetization is in the slowest mode with a few percent relaxing in higher modes.

In the slow diffusion regime, T_{2S0}^{-1} is constant with increasing κ and primarily depends on the self-diffusion coefficient,

$$T_{2S0}^{-1} = \frac{2\alpha D}{a^2}, \quad (3)$$

where α is a factor accounting for the geometry of the pore space (α is three for spherical pores) (Godefroy et al., 2001). The fastest relaxation mode (or initial surface relaxation rate) $T_{2S\infty}^{-1}$ is given by

$$T_{2S\infty}^{-1} = \rho_2 \frac{S_{\text{pore}}}{V_{\text{pore}}}, \quad (4)$$

where ρ_2 is the surface relaxivity. In the intermediate diffusion regime, the zeroth mode relaxation time is given by a combination of equations 3 and 4 (Godefroy et al., 2001). Figure 1 shows the behavior of the zeroth mode relaxation rate T_{2S0}^{-1} as a function of κ for water relaxing in a single spherical pore.

NMR relaxation in the fast-diffusion regime

For water in a single pore relaxing in the fast-diffusion regime the solution to the modified Bloch equation (equation 1) is a single exponential decay (i.e., all relaxation occurs in the zeroth mode),

$$M_{xy}(t) = M(0) \exp(-t/T_{2S}), \quad (5)$$

where T_{2S} is the surface relaxation time (T_{2S}^{-1} is the surface relaxation rate). As with equation 2, the term accounting for bulk relaxation, $\exp(-t/T_{2B})$, has been divided out of $M_{xy}(t)$ in equation 5. T_{2S}^{-1} is given by

$$T_{2S}^{-1} = \rho_2 \frac{S_{\text{pore}}}{V_{\text{pore}}}. \quad (6)$$

Comparing equation 6 to equation 4, we see that the surface relaxation rate in the fast-diffusion regime corresponds to the initial surface relaxation rate in the slow-diffusion regime. Figure 1 demonstrates the transition of the zeroth mode relaxation rate in the slow-diffusion regime to the surface relaxation rate in the fast-diffusion regime as a function of κ . The surface relaxivity term in equations 4 and 6 has been shown experimentally to be related to the surface concentration and mineralogy of paramagnetic species on the surface (Foley et al., 1996; Bryar et al., 2000; Keating and Knight, 2007, 2010).

NMR relaxation in porous media with a single surface relaxivity

The previous discussion focused on the NMR response of water in a single pore. Geologic material consists of multiple pores with multiple pore environments, where *pore environment* refers to the pore-scale conditions controlling NMR relaxation, i.e., the surface relaxivity and the surface-area-to-volume ratio. For a water-saturated geologic material, the measured NMR signal $I(t)$ is a multiple exponential decay,

$$I(t) = I(0) \sum_j h_j \exp(-t/T_{2j}), \quad (7)$$

where $I(0)$ is the total initial signal intensity and is proportional to the total water content detected by the NMR relaxation measurement,

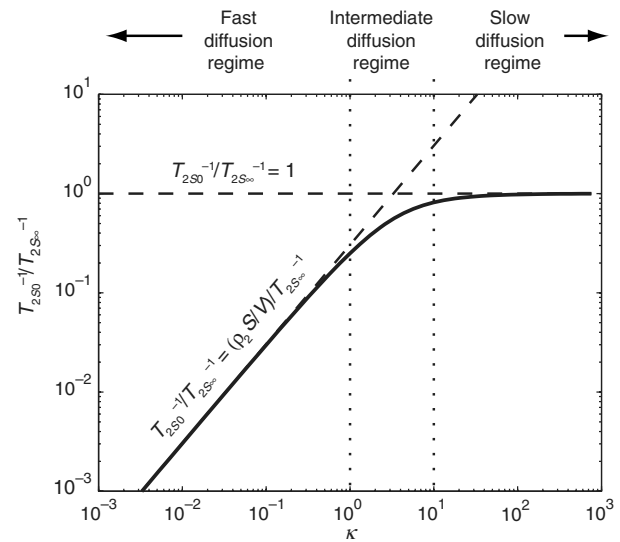


Figure 1. Behavior of T_{2S0}^{-1} versus κ for the fast-, intermediate-, and slow-diffusion regimes.

h_j is the normalized number of protons relaxing with a rate of T_{2j}^{-1} . We note that, unlike equations 2 and 5, the bulk fluid relaxation term is included in equation 7. In the fast-diffusion regime, neglecting the effect of magnetic field inhomogeneities and assuming that the protons in the measured sample either do not diffuse or only minimally diffuse from pore to pore (i.e., there is weak coupling), each pore environment is represented by a single exponential decay in equation 7 where the j th relaxation rate T_{2j}^{-1} is given by

$$T_{2j}^{-1} = T_{2B}^{-1} + (T_{2S})_j^{-1}, \quad (8)$$

where $(T_{2S})_j^{-1}$ is the surface relaxation rate of the j th pore environment and is given by equation 6. Additionally, h_j is often plotted versus T_{2j} , yielding a distribution of relaxation times. Under the assumption of fast diffusion, the relaxation time distribution represents the distribution of pore environments in the measured volume. Furthermore, when the surface relaxivity is uniform across all pores in the pore space, the relaxation time distribution can be directly related, through a linear transform, to the distribution of $V_{\text{pore}}/S_{\text{pore}}$ and generally is assumed to represent the pore-size distribution in the measured volume (e.g., Kleinberg et al., 1994; Arns, 2004).

The arithmetic mean of $\log T_{2j}$, T_{2ML} , is commonly taken to represent the NMR behavior of the measured volume,

$$\log T_{2ML} = \sum_j h_j \log T_{2j}. \quad (9)$$

T_{2ML} is described by

$$T_{2ML}^{-1} = T_{2B}^{-1} + T_{2S}^{-1}, \quad (10)$$

where T_{2S}^{-1} in equation 10 can be related to the surface-area-to-volume ratio of the total pore space S/V , assuming uniform surface relaxivity:

$$T_{2S}^{-1} = \rho_2 \frac{S}{V}, \quad (11)$$

where ρ_2 and S/V in the remainder of this paper now denote averages taken over the entire measured volume. In geologic materials, T_{2S}^{-1} is typically much greater than T_{2B}^{-1} , and therefore, $T_{2ML}^{-1} \approx T_{2S}^{-1}$. It is this relationship between T_{2ML}^{-1} and S/V that allows NMR measurements to be used to predict the hydraulic conductivity or permeability in water-saturated geologic material (e.g., Seevers, 1966; Mohnke and Yaramanci, 2008).

To test the assumption of fast diffusion, the control parameter needs to be calculated from D , ρ_2 , and a . Although D is known and ρ_2 can be determined from equation 11, it is not possible to directly determine a . The value of a is the average distance a proton needs to travel to reach a paramagnetic site; we use the average pore radius to represent this distance. A common approach for calculating a is to assume that $a = (S/V)^{-1}$ for the measured volume (e.g., Dunn et al., 2002). If the grains were perfectly smooth and spherical, this approach would give a value that is close to the pore radius and would serve as a good approximation for a ; however, in realistic geologic material where grains are rough and not spherical, using this approach to calculate a underestimates the pore radius (Keating and Knight, 2010). Instead, we use the approach outlined by Keating and Knight (2010) and calculate an upper limit on the

value of a using radius of the largest sphere that can be inscribed inside the pore space of a simple cubic packing of spherical grains, $a = 0.732r$, where r is the average radius of the grains (Guillot, 1982).

In the previous discussion, we have neglected a third term in equation 10, the diffusion relaxation rate, $T_{2D}^{-1} = D(\gamma G t_E)^2/12$, where G is the average magnetic field gradient, and t_E is the echo-time, an NMR measurement parameter (Kleinberg and Horsfield, 1990) that accounts for inhomogeneities in the magnetic field represented by ΔB_z in equation 1. The diffusion relaxation term is negligible when the static magnetic field is homogeneous, which is the case for laboratory measurements collected on material with low magnetic susceptibility (e.g., Keating and Knight, 2007). The validity of this assumption can be tested by determining T_{2ML}^{-1} for NMR data collected with multiple t_E . If T_{2ML}^{-1} remains constant with t_E , then magnetic field inhomogeneities are negligible.

NMR relaxation in porous media with spatially variable surface relaxivity

Our focus is the interpretation of NMR relaxation data, under conditions of fast diffusion, from samples containing surfaces with different surface relaxivities. Using equations 10 and 11 to represent the mean log relaxation rate assumes an average value of ρ_2 for the entire sample. In this section, we propose two relationships between the mean log relaxation rate and the surface relaxivity in the fast-diffusion regime for materials containing two surface types, each with a different associated value of ρ_2 . In the first relationship, it is assumed that there is weak coupling between pores, indicating that pore throats are small enough or the distance between pores with different pore environments is large enough that a proton only diffuses through a single pore environment over the time scale of the experiment. In the second relationship, we assume that there is strong coupling between pores meaning that pore throats are sufficiently large such that a single proton can move through and sample several pore environments over the time scale of the NMR experiment. We refer to the surface with the larger surface relaxivity value (i.e., the iron-coated surface) as the paramagnetic surface and the surface with the smaller surface relaxivity value (i.e., the quartz surface) as the nonparamagnetic surface. The surface relaxivity of the paramagnetic surface is denoted with ρ_{2P} and the surface relaxivity for the nonparamagnetic surface is denoted by ρ_{2N} .

The boundary between the region of weak coupling and region of strong coupling can be determined by comparing a (previously defined as the average distance between two paramagnetic sites) to the average distance a spin can diffuse through the pore space before relaxing, ℓ_D ; ℓ_D is calculated from Einstein's equation for self-diffusion, $\ell_D = \sqrt{6DT}$, where T is the relaxation time. As shown in a recent laboratory study, if $\ell_D < a$, then there is weak coupling; if $\ell_D > a$, then there is strong coupling (Grunewald and Knight, 2011).

Representing the response of a system with nonuniform surface relaxivity and weak coupling

In the case of weak coupling, protons have limited movement from pore to pore over the time scale of the experiment, and so each pore environment independently contributes to the relaxation time distribution. We assume that protons in a single pore environment only sample one type of surface during the NMR experiment and

define two regions in the pore space: one region where the relaxation rate is governed by the nonparamagnetic surface; the other where the relaxation rate is governed by the paramagnetic surface. In this case, we can equate the total measured signal to the sum of the signal due to relaxation governed by the nonparamagnetic surface, $I_N(t)$ and the signal due to relaxation governed by the paramagnetic surface $I_P(t)$:

$$I(t) = I_N(t) + I_P(t). \quad (12)$$

Expanding each term on the right-hand side using equation 7 gives

$$I(t) = I_N(0) \sum_{jN} h_{jN} \exp(-t/T_{2jN}) + I_P(0) \sum_{jP} h_{jP} \exp(-t/T_{2jP}), \quad (13)$$

where the subscript jN indicates the sum is taken over the pore environments where relaxation is governed by the nonparamagnetic surface, and the subscript jP indicates the sum is taken over the pore environments where relaxation is governed by the paramagnetic surface. The variable h_{jN} is the normalized number of protons relaxing at nonparamagnetic surfaces with relaxation time T_{2jN} (i.e., $0 \leq h_{jN} \leq 1$). The variable h_{jP} is the normalized number of protons relaxing at paramagnetic surfaces with relaxation time T_{2jP} . Because $I_N(0)$ is proportional to the volume of water in the sample affected by the paramagnetic surfaces, we can rewrite it as $I_N(0) = C_N I(0)$, where C_N is the volume fraction of water in the sample in which relaxation is governed by the nonparamagnetic surface. Similarly, we can write $I_P(0)$ as $I_P(0) = C_P I(0)$, where C_P is the fractional volume of water in the sample in which relaxation is governed by the paramagnetic surface. Substituting these equations for $I_N(0)$ and $I_P(0)$ into equation 13 and using the definition of the mean log relaxation time from equation 9, we obtain

$$\log T_{2ML} = \left(C_N \sum_{jN} h_{jN} \log(T_{2jN}) + C_P \sum_{jP} h_{jP} \log(T_{2jP}) \right) = C_N \log(T_{2ML-N}) + C_P \log(T_{2ML-P}), \quad (14)$$

where T_{2ML-N}^{-1} and T_{2ML-P}^{-1} are the mean log relaxation times for the protons with relaxation governed by the nonparamagnetic and paramagnetic surfaces, respectively. In the interpretation of our data, we assume that we can use the volume fraction of the coated grains C_P to represents the volume fraction of water in which relaxation is governed by the paramagnetic surfaces. Similarly, we also use the volume fraction of the uncoated grains C_N to represent the volume fraction of water in which relaxation is governed by nonparamagnetic surfaces.

We now expand T_{2ML-N}^{-1} and T_{2ML-P}^{-1} to recover the link between T_{2ML} , the single NMR parameter typically used to represent a measurement, and the surface relaxivity values ρ_{2P} and ρ_{2N} for the weakly coupled system. Because these expressions are derived assuming a weakly coupled (WC) system, we refer to the surface relaxivity values as ρ_{2P-WC} and ρ_{2N-WC} :

$$T_{2ML-P}^{-1} = T_{2B}^{-1} + \rho_{2P-WC} \frac{S_P}{V_P}, \quad (15a)$$

$$T_{2ML-N}^{-1} = T_{2B}^{-1} + \rho_{2N-WC} \frac{S_N}{V_N}, \quad (15b)$$

where S_P is the total surface area of the paramagnetic surfaces, V_P is the volume of water with relaxation governed by the paramagnetic surfaces, S_N is the total surface area of the nonparamagnetic surfaces, and V_N is the volume of water with relaxation governed by the nonparamagnetic surfaces. Here, we have assumed that the volume of water with relaxation governed by paramagnetic surfaces is independent of the volume of water with relaxation governed by the nonparamagnetic surfaces, that is, $V = V_N + V_P$.

Representing the response of a system with nonuniform surface relaxivity and strong coupling

For the case of strong coupling, protons can diffuse through multiple different pore environments over the time scale of the experiment and the relaxation rate is an average of all pore environments. [Foley et al. \(1996\)](#) define T_{2S}^{-1} for a single pore with magnetically active (i.e., high values of ρ_2) surface sites interspersed with non-active (i.e., very low values of ρ_2) surfaces sites. We use an equivalent expression to describe relaxation in a porous medium where there are paramagnetic surfaces interspersed with nonparamagnetic surfaces and the relaxation of diffusing protons is affected by both types of surfaces:

$$T_{2ML}^{-1} = T_{2B}^{-1} + T_{2S}^{-1} = T_{2B}^{-1} + \rho_{2P-SC} \frac{S_P}{V} + \rho_{2N-SC} \frac{S_N}{V}. \quad (16)$$

The subscript SC indicates that this is the surface relaxivity in a system with strong coupling.

Equations 15 and 16 are the two equations we used in the assessment of NMR data from samples with spatially variable surface relaxivity. The question we ask: Do these equations represent the relationship between the mean log relaxation time and the surface relaxivity in porous media with spatially variable surface relaxivity? Given that both equations have two surface relaxivity values, we used the known value of ρ_{2N} , from measurements on a homogeneous sample, for the values of ρ_{2N-WC} and ρ_{2N-SC} . This allowed us to then calculate the surface relaxivity of the paramagnetic surface, the surface that governs the NMR relaxation. Comparing the values of ρ_{2P-WC} or ρ_{2P-SC} determined in this manner with the value of the surface relaxivity calculated from a uniform sample allowed us to assess the validity of these equations.

Testing the fast-diffusion assumption in porous media with spatial variable ρ_2

The relationships between surface relaxivity and surface relaxation rate in any of equations 11, 15, and 16 are only valid in the fast-diffusion regime and require that the assumption of fast diffusion be verified. For samples with uniform surface relaxivity, ρ_2 is first determined from T_{2S}^{-1} and S/V using equation 11, and then the assumption of fast diffusion is tested by calculating κ from ρ_2 ($\kappa = \rho_2 a/D$). Because this approach assumes fast diffusion to calculate ρ_2 , if relaxation does not occur in the fast-diffusion regime, then the resulting value of ρ_2 , and consequently, κ , will be underestimated. To avoid misinterpreting the boundary between the fast- and intermediate-diffusion regimes, we thus use $\kappa < 0.1$ as

the condition for accepting the assumption of fast diffusion; this condition was shown to be sufficient by Ryu (2009). However, for samples with spatially varying surface relaxivity as used in this study, there are multiple surfaces with multiple surface relaxivity values, so a different approach for calculating κ must be used. For the samples in this study, we use the surface that governs relaxation, the paramagnetic surface, to calculate the control parameter. By calculating the control parameter using the surface relaxivity for the paramagnetic surface and the value of a as the average distance between two paramagnetic surfaces we obtain an upper bound on the control parameter. The control parameter for the case of weak coupling is denoted as κ_{P-WC} , where $\kappa_{P-WC} = \rho_{2P-WC}a/D$; the control parameter for the case of strong coupling is denoted as κ_{P-SC} , where $\kappa_{P-SC} = \rho_{2P-SC}a/D$. For equations 15 and 16, we accept the fast-diffusion assumption if $\kappa_{P-WC} < 0.1$ or $\kappa_{P-SC} < 0.1$, respectively.

LABORATORY EXPERIMENTS

We designed a set of experiments that would allow us (1) to evaluate the equations linking surface relaxivity to NMR relaxation rates in samples with spatially variable surface relaxivity and (2) to assess the impact of assuming uniform ρ_2 when samples have spatially varying ρ_2 . We worked with uniform samples of quartz and iron-coated quartz grains, and with spatially variable samples prepared by mixing the two types of grains. In our data analysis, we started by first calculating the average ρ_2 for all samples using equation 11 ($\rho_2 = T_{2S}^{-1}/[S/V]$). We then compared the ρ_2 values from the uniform samples (with single-valued ρ_2) to those from the spatially variable samples. Next, we evaluated the validity of the two equations that we have proposed for samples with spatially variable ρ_2 by using equations 15 and 16 to calculate the apparent surface relaxivity of the paramagnetic surface assuming weak coupling, ρ_{2P-WC} , and strong coupling, ρ_{2P-SC} . The determined values for surface relaxivity were compared with the known values from the uniform samples. Finally, we tested the assumption of fast diffusion for each surface relaxivity by calculating the control parameter κ , κ_{P-WC} , or κ_{P-SC} .

Sample preparation

To create samples with spatially varying ρ_2 , mixtures were prepared that contained two materials known to have different values of ρ_2 : quartz sand and quartz sand coated with an iron-mineral. The quartz sand Si(IV)O₂; IOTA Standard, Unimin Corporation, used in this study was chosen for its low concentration of paramagnetics (<0.1 ppm Fe) and its uniform grain size (74 to 297 μ m in diameter). Two iron-(hydr)oxide minerals were used to coat the quartz sand: ferrihydrite (Fe(OH)₃ · nH₂O) and hematite (Fe₂O₃). To ensure that the surface relaxivity of the sands coated with ferrihydrite was different from the surface relaxivity of the sands coated with hematite, a different concentration of iron was used in each coating. The quartz surface is the nonparamagnetic surface, whereas the iron-coated surface is the paramagnetic surface. Given what was found in our earlier work (Keating and Knight, 2007), we expect that the methods used to coat the quartz sands with ferrihydrite or hematite will produce samples with surface distributions of iron minerals that are similar, independent of mineral type.

To prepare the ferrihydrite-coated sands, ferrihydrite was first synthesized by the titration of Fe(NO₃)₃ · H₂O (0.4 M) with

1 M NaOH to pH 7 (Hansel et al., 2003; Keating and Knight, 2007). Once synthesized, the resulting ferrihydrite slurry was washed by centrifuging to obtain a ferrihydrite pellet, decanting excess water, then resuspending the ferrihydrite pellet in distilled, deionized water. This process was repeated three times to remove excess salts. The ferrihydrite slurry was then mixed with quartz sand to obtain a total iron concentration of less than 0.5% iron by weight (g-Fe/g-sand) and left to dry. Once dry, the coated sand was rinsed with deionized water three times to remove any ferrihydrite not attached to the mineral surface.

To prepare the hematite-coated sands, hematite was first synthesized by the forced hydrolysis of a Fe(III) salt solution (1 M Fe(NO₃)₃) (Schwertmann and Cornell, 2000; Keating and Knight, 2007). Once synthesized, the hematite slurry was filtered by dialysis to remove excess salts from the slurry. The hematite slurry was then mixed with quartz sand to obtain a total iron concentration greater than 1% iron by weight and left to dry. Once dry, the coated sand was rinsed with deionized water three times to remove any hematite not attached to the mineral surface.

Samples with spatial variation in ρ_2 were created by mixing each iron-coated sand with pure quartz sand to obtain concentrations by weight C_P of 1%, 2%, 5%, 10%, 25%, 50%, and 75% in grams of iron-coated sand to total grams of sand. Additionally, samples with uniform ρ_2 also were created; these samples contained pure quartz sand ($C_P = 0\%$) and pure iron-coated sand ($C_P = 100\%$).

NMR measurements

To create the NMR samples, each mixture was packed into a cylindrical Teflon sample holder (height 56.3 mm, inner diameter 31.0 mm). The lid of each sample holder was perforated with holes less than 0.1 mm in diameter; a filter (25 μ m) was placed at the top of the sample to prevent sand grains from escaping from the sample holder during the saturation process. Once packed, the NMR samples were weighed, and the mass of each sample was determined.

Each NMR sample was saturated by placing the sample, submerged in a beaker of distilled deionized water (pH ~ 7), in a vacuum chamber and reducing the pressure in the chamber to 100 mmHg for 30 min. The volume of water saturating the sample was then determined gravimetrically. Once saturated, the sample holders were covered with Parafilm to prevent water in the sample from evaporating over the course of the NMR experiment.

NMR relaxation data were collected with a 2.2 MHz Maran Ultra NMR Core Analyzer (Resonance Instruments) using a CPMG (Carr-Purcell-Meiboom-Gill) pulse sequence. One data point was obtained at each echo in the CPMG pulse sequence; 20,000 echoes were used for all samples. Data were collected at four echo times, $t_E = 300, 400, 600,$ and 800μ s with an instrument dead time of 60 μ s and stacked 52 times. The measurements at each echo time were repeated three times. Measurements were consistently made at 30°C, which is the standard operating temperature of the instrument. Following the NMR measurements, each sample was reweighed to determine if any water had evaporated from the sample over the course the NMR experiment.

Once the NMR data had been collected, the pore water was removed from each sample by vacuum filtration. The bulk fluid relaxation rate was measured using the extracted pore water at a single echo time of 300 μ s; 32,000 echoes were used for the pore water measurements. All other measurement parameters were the same as for the NMR samples.

The relaxation time distribution for each NMR data set was determined using a nonnegative least-squares inversion with Tikhonov regularization as described by Whittall et al. (1991). Prior to applying the algorithm, each NMR data set was logarithmically subsampled to 5000 data points to improve the speed of the fitting algorithm. The subsampled data were then fit to a distribution of 160 exponentially spaced T_2 -values ranging from 1 ms to 5 s. The regularization parameter was selected such that each datum was misfit by approximately one standard deviation.

Sample characterization

One subsample of approximately 10 g was taken from the 0% and 100% iron-coated sand samples for surface-area analysis. The specific surface area, defined as the surface area normalized by the mass of the sample, was measured for each of these samples using the Brunauer-Emmett-Teller (BET) adsorption method with $N_2(g)$ as the adsorbate (measurements performed by Micromeritics Analytical Services). The measurements were collected with the Micromeritics Tristar II 3020 Surface Area Analyzer, which produces accurate results (within $\sim 1\%$ of the measured surface area) for samples with surface areas as low as $0.01 \text{ m}^2/\text{g}$.

RESULTS AND DISCUSSION

Physical properties

The volume of the pore space V was taken to be the average volume of water in the saturated sample determined gravimetrically before and after the NMR measurement; error in V was calculated from the standard deviation from two measured volumes. This resulted in V ranging from $18.26 \pm 0.01 \text{ mL}$ for the quartz sample to $19.36 \pm 0.01 \text{ mL}$ for the sample containing 10% ferrihydrite-coated sand and did not show a trend with iron concentration or mineral type.

For each sample, the porosity ϕ , given for the pure quartz sand and the 100% iron-coated sand samples in Table 1, was calculated from the volume of water in the sample divided by the volume of the sample holder. Errors on the porosity were calculated using error propagation. The porosity ranged from 0.432 ± 0.001 for the pure quartz sand sample to 0.458 ± 0.001 for the mixtures containing 10% ferrihydrite-coated sand. The porosity did not show any trend with iron-coated sand concentration.

The measured specific surface areas for the samples with uniform surface relaxivity are given in Table 1; the errors given are the errors associated with the multipoint BET measurement technique. The measured specific surface area for pure quartz sand S_{sN} was $0.1136 \pm 0.0005 \text{ } \mu\text{m}^2/\text{g}$, which is close to the value found for a similar quartz sand in Keating and Knight (2007). The measured specific surface area for the 100% iron-coated sands S_{sP} was $1.1993 \pm 0.003 \text{ } \mu\text{m}^2/\text{g}$ for ferrihydrite-coated sand and $1.058 \pm 0.002 \text{ } \mu\text{m}^2/\text{g}$ for hematite coated-sand. A large increase in the surface area due to the addition of the iron coating is expected due to the high surface areas of the iron minerals. For pure ferrihydrite (i.e., ferrihydrite not coated to the surface of quartz), the expected surface area ranges from 200 to $400 \text{ m}^2/\text{g}$ (e.g., Das et al., 2011); for the hematite synthesized in this study, the expected surface area is between 60 and $85 \text{ m}^2/\text{g}$ (Schwertmann and Cornell, 2000).

The measured values of S_{sP} for 100% iron-coated sands were not the same as the values determined in Keating and Knight (2007), as expected due to the different concentrations of iron used in each of the coatings.

The measured values S_{sN} and S_{sP} also were used to calculate the ratio of the surface area of the quartz sand to the pore volume and the ratio of the surface area of the iron-coated sand to the pore volume ratio, S_N/V and S_P/V , respectively, for each sample from

$$\frac{S_i}{V} = mC_i \frac{S_{si}}{V}, \quad (17)$$

where i indicates either quartz sand (N) or iron-coated sand (P), and m is the mass of the solid phase within the sample. For the ferrihydrite-coated sand mixtures and the hematite-coated sand mixtures, S_N/V ranged from $0 \text{ } \mu\text{m}^{-1}$ for the sample with $C_P = 100\%$ to $0.393 \pm 0.002 \text{ } \mu\text{m}^{-1}$ for the sample with $C_P = 0\%$. For the ferrihydrite-coated sand mixtures, S_P/V ranged from $0 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 0\%$ to $3.93 \pm 0.02 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 100\%$; for the hematite-coated sand mixtures, S_P/V ranged from $0 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 0\%$ to $3.657 \pm 0.009 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 100\%$. For all iron-coated sand mixtures, S_N/V decreased with iron-coated sand concentration and S_P/V increased with iron-coated sand concentration.

The sum of S_N/V and S_P/V was used to calculate the total surface-area-to-volume ratio S/V for all samples. The values of S/V for the 0% and 100% iron-coated sand samples are given in Table 1. (For the 0% iron-coated sand samples, the value of S/V is equivalent to S_N/V ; and for the 100% iron-coated sand samples, the value of S/V is equivalent to S_P/V .)

For ferrihydrite-coated sand samples containing multiple surface types, S/V ranged from $0.432 \pm 0.002 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 1\%$ to $2.696 \pm 0.006 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 75\%$. For hematite-coated sand samples containing multiple surface types, S/V ranged from $0.416 \pm 0.002 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 1\%$ to $3.06 \pm 0.01 \text{ } \mu\text{m}^{-1}$ for the mixture with $C_P = 75\%$.

Relaxation time distributions

Figure 2 shows the T_2 -distributions at an echo time of $300 \text{ } \mu\text{s}$ for the ferrihydrite-coated sand mixtures (Figure 2a) and the hematite-coated sand mixtures (Figure 2b) for C_P equal to 0, 2, 10, 50, and 100%.

The shape of the relaxation time distributions for the samples with uniform surface relaxivity compared well with the shape of the relaxation time distributions found for similar samples in Keating and Knight (2007). The T_2 -distribution for the pure quartz sand sample was a single peak centered at a long relaxation times. For both of the 100% iron-coated sand samples, the T_2 -distributions had

Table 1. The physical properties of the measured samples.

Sample type	C_P (%)	S_s (m^2/g)	ϕ	S/V (μm^{-1})
Quartz sand	0	0.1136 ± 0.0005	0.432 ± 0.001	0.393 ± 0.002
Ferrihydrite-coated sand	100	1.199 ± 0.003	0.448 ± 0.002	3.93 ± 0.02
Hematite-coated sand	100	1.058 ± 0.002	0.439 ± 0.001	3.657 ± 0.009

two peaks: a large, broad peak centered at long relaxation times and a smaller, broad peak centered at short relaxation times.

For all iron-coated sand mixtures, we found that the shape of the T_2 -distribution depended on C_P . When C_P was less than 10%, the distributions had a single narrow peak centered at long relaxation

times close to, but less than, the central relaxation time for the pure quartz sand sample. When C_P increased above 10%, the peak at long relaxation times broadened and shifted toward shorter relaxation times. Additionally, a second peak developed at short relaxation times that broadened as C_P increased to 100%. Similar trends for T_2 -distributions were seen for mixtures containing different ratios of quartz and hematite grains (Grunewald and Knight, 2011).

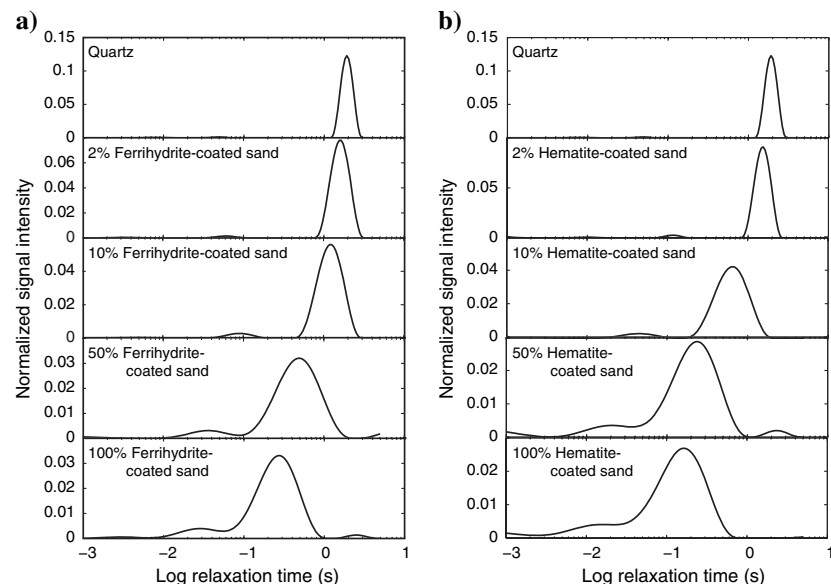


Figure 2. Relaxation time distributions, plotted as the log relaxation time versus the normalized signal intensity, for the 2%, 10%, 50%, and 100% ferrihydrite-coated sand samples (a) and the hematite-coated sand samples (b). Also shown for reference on each figure is the normalized relaxation time distribution for the pure quartz sand sample. All relaxation time distributions shown were obtained at an echo time of 300 μ s.

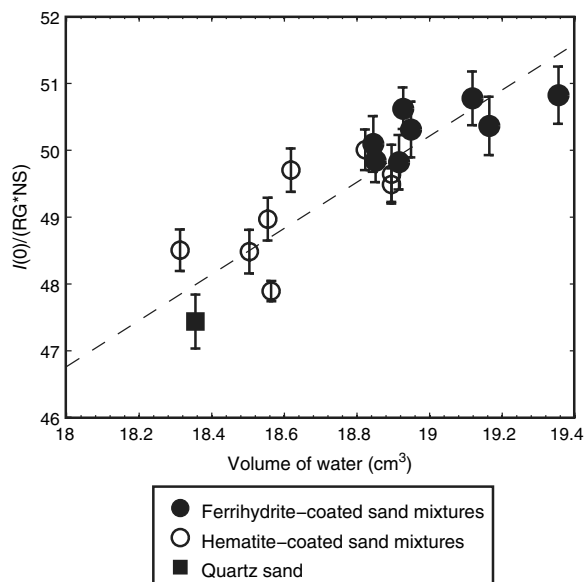


Figure 3. The sum of the total initial signal intensity $I(0)$ scaled by the receiver gain RG and the number of scans NS versus the volume of water for all mixture types for the data collected at an echo time of 300 μ s. The dashed line represents a linear fit to the data. Error bars were determined from the standard deviation of repeated measurements; error bars not shown are smaller than the size of the data points.

Total signal intensity

The total initial signal intensity $I(0)$ was calculated for each sample. We can use $I(0)$ to determine if the total water content is detected in the relaxation time distribution curves. Although $I(0)$ is proportional to the total water content, its magnitude is affected by the receiver gain RG and the number of scans per sample NS . Figure 3 shows the plot of $I(0)$ at an echo time of 300 μ s, scaled by $RG \times NS$, versus the volume of water V for each sample. As expected, the scaled signal intensity increased with water volume and did not show any trend with iron-mineral type or iron-mineral concentration. Although the water volume was expected to be directly proportional to scaled signal intensity, some variation on the order of 2% from the expected value was observed in the data likely due to capillary- or surface-bound measurement.

Mean log relaxation rate

The T_2 -distributions were used to determine T_{2ML}^{-1} from equation 9 at each t_E for the measured samples. The values of T_{2ML}^{-1} at an echo time of 300 μ s are plotted versus C_P in Figure 4. Figure 4a contains the results from the ferrihydrite-coated sands; Figure 4b contains the results from the hematite-coated sands. The values of T_{2ML}^{-1} plotted in Figure 4 are the average of the value determined from the three measurements collected on each sample. The error on T_{2ML}^{-1} is the standard deviation; error bars not shown on Figure 4 are smaller than the size of the data points. The average value of T_{2ML}^{-1} for the pure quartz sand sample at an echo time of 300 μ s was 0.55 ± 0.01 s $^{-1}$. The average value of T_{2ML}^{-1} at an echo time of 300 μ s for the 100% ferrihydrite-coated sand sample was 5.2 ± 0.1 s $^{-1}$ and for the 100% hematite coated sand sample was 10.8 ± 0.2 s $^{-1}$. For the ferrihydrite- and hematite-coated sand mixtures, the value of T_{2ML}^{-1} increased from pure quartz sand to that of the 100% iron-coated sands with increase in C_P .

For all samples, T_{2ML}^{-1} did not show a linear increase with t_E^2 , which implies that T_{2D}^{-1} is negligible for the samples measured in this study. This same result was found in the previous study of Keating and Knight (2007) where, in samples of ferrihydrite-coated sand and hematite-coated sand, T_{2D}^{-1} was found to be zero.

Bulk fluid relaxation rate

The bulk fluid relaxation rate was determined for each sample type using the NMR relaxation measurements of the extracted fluid. The calculated values of T_{2B}^{-1} versus C_P are included in Figure 4.

The errors are the standard deviation calculated using the results from multiple measurements; error bars not shown are smaller than the size of the data points. The value of T_{2B}^{-1} for the pure quartz sand sample was $0.42 \pm 0.02 \text{ s}^{-1}$. For the 100% ferrihydrite-coated sand sample, T_{2B}^{-1} was $0.60 \pm 0.02 \text{ s}^{-1}$; for the 100% hematite-coated sand sample, T_{2B}^{-1} was $1.1 \pm 0.1 \text{ s}^{-1}$. For the ferrihydrite-coated sand mixtures, T_{2B}^{-1} had a minimum value of $0.37 \pm 0.02 \text{ s}^{-1}$ for $C_P = 10\%$ and a maximum value of $0.58 \pm 0.01 \text{ s}^{-1}$ for $C_P = 1\%$. For the hematite-coated sand mixtures, T_{2B}^{-1} had a minimum value of $0.40 \pm 0.02 \text{ s}^{-1}$ for $C_P = 10\%$ and a maximum value of $0.91 \pm 0.06 \text{ s}^{-1}$ for $C_P = 75\%$. The fact that T_{2B}^{-1} for any sample containing iron-coated sand was greater than T_{2B}^{-1} for the pure quartz sand sample is most likely due to an increase in the concentration of hematite or ferrihydrite particles suspended in solution. This cannot be attributed to an increase in the concentration of dissolved Fe(III) in the measured fluids because ferrihydrite and hematite have low solubility in water at pH 7 (Cornell and Schwertmann, 2003).

Surface relaxation rate

The value of T_{2S}^{-1} was determined from the calculated values of T_{2ML}^{-1} and T_{2B}^{-1} measured at an echo time of $300 \mu\text{s}$ using equation 10. The calculated values of T_{2S}^{-1} are plotted versus concentration in Figure 4. The errors are the standard deviation calculated using the results from multiple measurements; error bars not shown are smaller than the size of the data points. The value of T_{2S}^{-1} for the pure quartz sand sample was $0.11 \pm 0.02 \text{ s}^{-1}$. For the 100% ferrihydrite-coated sand sample, T_{2S}^{-1} was $5.1 \pm 0.5 \text{ s}^{-1}$; for the 100% hematite-coated sand sample, T_{2S}^{-1} was $10.1 \pm 0.3 \text{ s}^{-1}$. For the ferrihydrite-coated sand mixtures, T_{2S}^{-1} ranged from $0.15 \pm 0.02 \text{ s}^{-1}$ for $C_P = 1\%$ to $3.8 \pm 0.4 \text{ s}^{-1}$ for $C_P = 75\%$. For the hematite-coated sand mixtures, T_{2S}^{-1} ranged from $0.22 \pm 0.02 \text{ s}^{-1}$ for $C_P = 1\%$ to $8.4 \pm 0.4 \text{ s}^{-1}$ for $C_P = 75\%$.

The relationship between the three calculated relaxation rates and iron-coated sand concentration (Figure 4) showed the following: (1) T_{2ML}^{-1} and T_{2S}^{-1} increased with C_P . However, (2) T_{2B}^{-1} had no clear trend with C_P . (3) For the samples with $C_P > 5\%$ iron-coated sand, $T_{2S}^{-1} > T_{2B}^{-1}$; for the samples with $C_P \leq 5\%$, $T_{2B}^{-1} > T_{2S}^{-1}$.

Calculation of surface relaxivity

Assumption of uniform ρ_2

We first calculated ρ_2 of all of our samples, assuming that the surface is uniform using equation 11, $\rho_2 = T_{2S}^{-1}/(S/V)$. This equation returns a single value of relaxivity to represent the sample. We use ρ_{2Q} to refer to the surface relaxivity for the pure quartz sand sample, and found $\rho_{2Q} = 0.28 \pm 0.06 \mu\text{m/s}$. The surface relaxivity for the 100% ferrihydrite-coated sand sample ρ_{2F} was $1.3 \pm 0.1 \mu\text{m/s}$. The surface relaxivity for the 100% hematite-coated sand sample ρ_{2H}

was $2.8 \pm 0.1 \mu\text{m/s}$. The errors were calculated from error-propagation analysis.

The single-surface relaxivity values determined for the samples with spatial variation in surface relaxivity are shown in Figure 5 as a function of C_P . For the ferrihydrite-coated sand mixtures (Figure 5a) and the hematite-coated sand mixtures (Figure 5b), ρ_2 increased steeply with iron-coated sand concentration up to $C_P = 10\%$, then roughly plateaued with some variability at higher levels of C_P .

Accounting for spatially variable ρ_2

We next calculated the surface relaxivity of the paramagnetic surfaces using the equations presented earlier for systems with spatially variable ρ_2 : (1) We calculated the surface relaxivity of the paramagnetic surfaces assuming weak coupling, ρ_{2P-WC} , using equations 14 and 15. (2) We calculated the surface relaxivity of the paramagnetic surfaces assuming strong coupling, ρ_{2P-SC} , using equation 16. The plots of ρ_{2P-WC} and ρ_{2P-SC} versus C_P are shown in Figure 6 for the ferrihydrite-coated sand mixtures (Figure 6a) and for the

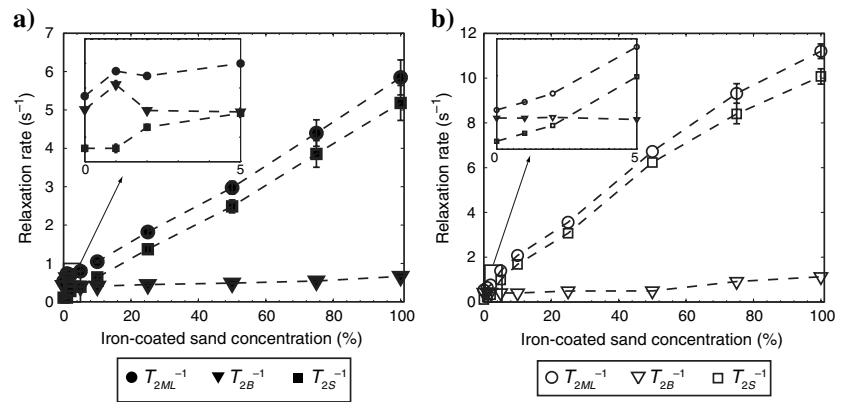


Figure 4. Values of T_{2ML}^{-1} , T_{2B}^{-1} , and T_{2S}^{-1} versus the iron-coated sand concentration for the measurements collected at $t_E = 300 \mu\text{s}$. Results are shown for ferrihydrite-coated sand (a) and hematite-coated sand (b). Error bars were calculated from the standard deviations of three repeated measurements; error bars not shown are smaller than the size of the data point.

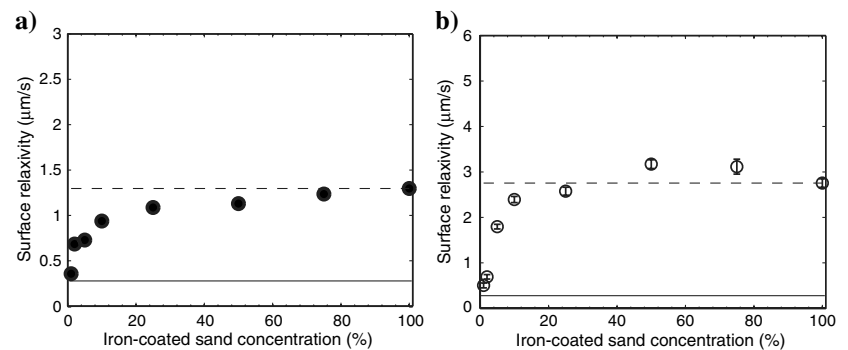


Figure 5. The average surface relaxivity for each sample ρ_2 versus iron-coated sand concentration C_P . Results are shown for ferrihydrite-coated sand (a) and hematite-coated sand (b). Error bars were calculated from error propagation; error bars not shown are smaller than the size of the data point. Also shown for reference in each graph is a dashed line representing the value of ρ_{2P} for the uniform iron-coated samples, ρ_{2F} for the ferrihydrite-coated sand, and ρ_{2H} for the hematite-coated sand; and a solid line representing the value of ρ_{2N} for the uncoated quartz sand surface.

hematite-coated sand mixtures (Figure 6b). Also shown as dashed lines on Figure 6 are ρ_{2F} for the 100% ferrihydrite-coated sand sample and ρ_{2H} for the 100% hematite coated-sand sample, both calculated using equation 11.

To determine the boundary between the regions of weak and strong coupling, we compared ℓ_D to a . An upper limit on the value of ℓ_D was determined by using $T = T_{2ML}$ for the pure quartz sand sample. This approach yielded a value for ℓ_D of 165 μm .

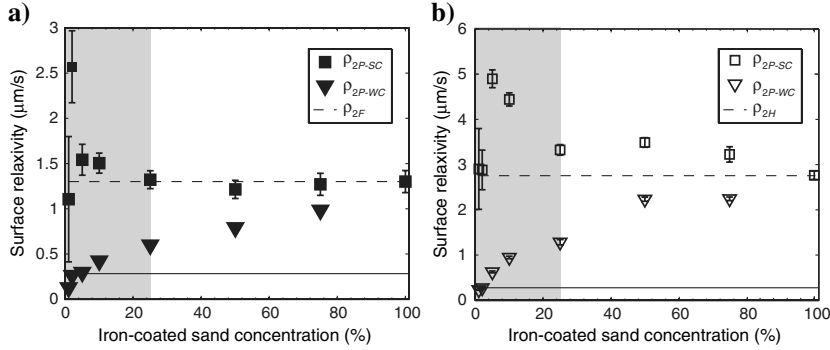


Figure 6. The values of ρ_{2P-SC} , ρ_{2P-WC} versus C_P for the ferrihydrite-coated sand mixtures (a) and the hematite-coated sand mixtures (b). Error bars were calculated from error propagation; error bars not shown were smaller than the size of the data point. Also shown for reference in each graph is a dashed line representing the value of ρ_{2P} for the uniform iron-coated samples, ρ_{2F} for the ferrihydrite-coated sand, and ρ_{2H} for the hematite-coated sand; and a solid line representing the value of ρ_{2N} for the uncoated quartz sand surface. The shaded area corresponds to the region of weak coupling.

Table 2. The values of a , κ , κ_{P-WC} , and κ_{P-SC} for the pure quartz sand, the ferrihydrite-coated sand mixtures, and the hematite-coated sand mixtures.

Mixture type	C_P (%)	a (μm)	κ	κ_{P-WC}	κ_{P-SC}
Quartz	0	82	0.01	—	—
Ferrihydrite	1	542	0.08	0.03	—
	2	409	0.11	0.04	—
	5	270	0.08	0.03	—
	10	193	0.07	0.03	—
	25	112	0.05	—	0.06
	50	82	0.04	—	0.04
	75	82	0.04	—	0.04
	100	82	0.04	—	—
Hematite	1	542	0.11	0.05	—
	2	409	0.11	0.04	—
	5	270	0.19	0.07	—
	10	193	0.18	0.07	—
	25	112	0.11	—	0.15
	50	82	0.10	—	0.11
	75	82	0.10	—	0.11
	100	82	0.09	—	—

Calculating the value of a is more challenging. An upper bound on the value of a , the average distance that a proton must travel to reach a paramagnetic site, was calculated using the average distance between the surfaces of two iron-coated grains assuming a simple cubic periodic packing structure. For mixtures with $C_P > 25\%$, a simple cubic packing structure results in at least one iron-coated grain in each pore. The value of a is then the radius of the largest sphere that can be inscribed inside the pore space of a simple cubic packing of spherical grains, which is $a = 0.732r$, where $r = 112 \mu\text{m}$, the radius of the quartz grains, which we assume to be very close to the radius of the coated grains. Using this approach, we found that for $C_P > 25\%$, $a = 82 \mu\text{m}$.

For the mixtures with $C_P \leq 25\%$, then we estimated a following the approach of Grunewald and Knight (2011), where an expression was derived to calculate the distance between two hematite grain surfaces in a mixture containing hematite and quartz sand grains. In this approach, the distance between the centers of two iron-coated grains d is given by

$$d = \left(\frac{4\pi}{2C_M(1-\phi)} \right)^{1/3} r. \quad (18)$$

The average distance that a proton would need to travel to reach a paramagnetic surface is then given by

$$a = \frac{1.28d}{2} - r. \quad (19)$$

The values of a estimated using these approaches are given in Table 2 for each iron-coated sand mixture and ranged from 82 to 542 μm . Comparing the values of a to ℓ_D , we found that weak coupling (i.e., $\ell_D \leq 25\%$) occurs for mixtures with $C_P \leq 25\%$ and strong coupling (i.e., $\ell_D > a$) occurs for mixtures with $C_P > 25\%$. The shaded region in Figure 6 is the weak-coupling region.

To evaluate the validity of the equations 15 and 16 to represent systems with spatially variable surface relaxivity, we compared each calculated value of ρ_{2P} (ρ_{2P-WC} in the weak-coupling region, ρ_{2P-SC} in the strong-coupling region) with the value of ρ_{2P} determined from the samples containing 100% iron-coated sand. We found (Figure 6a) excellent agreement, within 7%, in the strong-coupling region, between ρ_{2P-SC} and ρ_{2P} for the ferrihydrite-coated sand. The agreement was not as good, but within $\sim 26\%$, between ρ_{2P-SC} and ρ_{2P} for the hematite-coated sand (Figure 6b). This suggests that the equations used to represent a system with spatially varying ρ_2 and strong coupling are capturing the essential elements of the relaxation behavior.

In contrast, we did a very poor job of obtaining ρ_{2P-WC} for both systems in the region of weak coupling. The equation that we have used to represent the weak-coupling region assumes that the measured NMR signal is a sum of the NMR signal associated with relaxation at iron-coated surfaces and the signal associated with relaxation at quartz surfaces. That is, this equation assumes that a proton will only sample one type of surface during relaxation. In fact, this is unlikely to be the case even if the conditions for weak

coupling are satisfied. Weak coupling means that there is no diffusion between pores, but does not ensure that there is only a single type of surface within a pore and that diffusion does not occur between these different surfaces within a pore. It is most likely that in the weak-coupling region, the relaxation rate should be described by a combination of equations 15 and 16.

To determine the surface relaxivity values reported in this study, we have assumed that the surface area measured using nitrogen BET represents the surface area relevant for the NMR measurements. Although nitrogen BET measurements are commonly accepted as a means of determining the surface area of geologic materials, the measurement represents the surface area measured through the adsorption of nitrogen molecules, whereas the area of relevance for NMR studies is that of the water-wetted surface. The use of water as the adsorbate for surface area measurements has never been adopted for standard practice due to the additional complications involved in data acquisition and analysis. For example, the hydrogen bonding between water molecules results in clustering of the water on the surface, thus violating some of the underlying assumptions in the derivation of the BET equation. It is possible that the nitrogen-BET measured surface area is a lower bound on the water-wetted surface area, water being a smaller molecule and thus able to access more of a rough, natural surface. Conversely, it is possible that the surface area that is relevant for NMR measurements is a smoother surface area that more closely represents the average pore radius than the surface measurement by nitrogen BET; in this circumstance, the nitrogen-BET measurement would overpredict the relevant surface area. Either underpredicting or overpredicting the total surface area using nitrogen BET would mean that the surface relaxivities reported in this study are either consistently overpredicted or consistently underpredicted; however, we expect that the relationships between the surface relaxivities and the concentration of iron-coated sand would remain the same.

Assessing the validity of the fast diffusion assumption

In all cases, our calculations of surface relaxivity have assumed that relaxation was occurring in the fast-diffusion regime. We tested the assumption of fast diffusion by calculating the control parameter. We use κ to represent the control parameter calculated using the uniform or average surface relaxivity values, κ_{P-WC} for the control parameter calculated from ρ_{2P-WC} , and κ_{P-SC} for the control parameter calculated from ρ_{2P-SC} . The values of κ , κ_{P-WC} , and κ_{P-SC} are given in Table 2.

For the samples with uniform surface relaxivity, κ was 0.01 for the pure quartz sand, 0.04 for the ferrihydrite sand, and 0.09 for the hematite-coated sand. Using the guideline that κ must be < 0.1 to ensure fast diffusion, we conclude that, for the samples with uniform surface relaxivity, relaxation occurs in the fast-diffusion regime.

Next, we consider κ for the iron-coated sand mixtures in which ρ_2 was calculated using equation 11, i.e., assuming uniform ρ_2 . For the ferrihydrite-coated sand mixtures, κ was less than 0.1 (i.e., relaxation is in the fast-diffusion regime) for all the samples except for the sample with $C_P = 2\%$; for this sample, $\kappa = 0.11$. For the hematite-coated sand mixtures, $\kappa \geq 0.1$ for all concentrations and ranges from 0.10 to 0.19.

Next, we consider κ_{P-WC} for the iron-coated sand mixtures calculated using ρ_{2P-WC} ; because we calculated the control parameter assuming weak coupling, only samples that satisfy the weak coupling requirement, i.e., $C_P \leq 25\%$, are considered. For the

ferrihydrite-coated sand mixtures and the hematite-coated sand mixtures, we found that $\kappa_{P-WC} < 0.1$ for all samples with $C_P \leq 25\%$. For all samples, κ_{P-WC} indicated that relaxation occurred in the fast-diffusion regime.

From Figure 6, we see that ρ_{2P} is poorly predicted by ρ_{2P-WC} for the ferrihydrite- and hematite-coated sand mixtures in the region of weak coupling ($C_P \leq 25\%$). Because κ_{P-WC} was less than 0.1 in this region, this discrepancy does not at first appear to be due to a violation of the fast diffusion assumption; however, we note that because ρ_{2P-WC} underpredicted ρ_{2P} in this region, another approach would be to use $\kappa_P = \rho_{2P} a/D$, where ρ_{2P} is the surface relaxivity of the sample containing 100% iron-coated sand. In calculating κ_P for ferrihydrite- and hematite-coated sand mixtures, we found $\kappa_P \geq 0.1$ for all samples in the region of weak coupling indicating that the discrepancy between ρ_{2P} and ρ_{2P-SC} may arise because relaxation did not occur in the fast-diffusion regime. We note that because $\kappa_P \geq 0.1$ for ferrihydrite- and hematite-coated sands, the results presented here are not conclusive. An alternate explanation for the discrepancy between ρ_{2P} and ρ_{2P-SC} is that the volume of water used to represent each pore environment in equation 15 was not accurate.

Next, we consider κ_{P-SC} for the iron-coated sand mixtures calculated using ρ_{2P-SC} , respectively; here, only samples that are in the strong-coupling region ($C_P > 25\%$) are considered. For the ferrihydrite-coated sand mixtures, we found that $\kappa_{P-SC} < 0.1$; for the hematite-coated sand mixtures, we found that κ_{P-SC} was slightly greater than 0.1 for all mixtures and ranged from 0.11 to 0.15. As suggested by the good agreement between ρ_{2P-SC} and ρ_{2F} , the values of $\kappa_{P-SC} < 0.1$ for the ferrihydrite-coated sand mixtures lends further evidence to support that equation 16 is correct in the strong-coupling region. The value of κ_{P-SC} was slightly greater than 0.1 (between 0.11 and 0.15) for the hematite-coated sand mixtures; the fact that κ_{P-SC} is not strictly in the fast-diffusion regime might explain the discrepancy between ρ_{2P-SC} and ρ_{2H} for the hematite-coated sand mixtures.

CONCLUSIONS

It is generally assumed, in the analysis and interpretation of NMR data, that the surface relaxivity ρ_2 can be represented by a single value. Given the geochemical controls on ρ_2 , it is highly unlikely that all surfaces in a geologic material will have the same value of ρ_2 , highlighting a need to better understand systems with spatial variation in ρ_2 . This laboratory study has allowed us to explore simple systems created so as to have spatially varying ρ_2 . For samples with two types of grains, so assumed to have two different ρ_2 values, we found that we can use a relatively simple expression to describe relaxation in the strong-coupling region, where there is averaging over the different pore environments during the relaxation process. When the relaxation occurred in the weak-coupling region, we used an equation to represent the relaxation that presumes that the measured NMR signal is a sum of the signals associated with relaxation at the surfaces coming from regions with different values. This expression failed to capture what was observed in the data, which is likely because it is not possible to quantify the distinct volumes of water interacting with and relaxing at each surface-type.

Continued research improving the relationship between the NMR relaxation rates and the surface relaxivity will ultimately help to improve the interpretation NMR measurements in complex, geologic material with nonuniform surface relaxivity.

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APPENDIX A

Summary of mathematical symbols.

Symbol	Definition
$M_{xy}(t)$	Transverse magnetization as a function of time t
$M_{xy}(\mathbf{x}, t)$	Vector representation of transverse magnetization as a function of space $\mathbf{x}$ and time t .
a	Mean distance that a proton must travel to reach a paramagnetic site
α	Factor accounting for the geometry of the pore space
f_k	Relative amplitude for each exponent relaxing in the k th relaxation mode
G	Average magnetic field gradient
t_E	Echo-time
ℓ_D	Average distance a spin can diffuse through the pore space before relaxing
$I(t)$	Measured NMR signal
h_j	Fraction of water relaxing in the j th pore environment
h_{jP}	Fraction of water relaxing in the j Pth pore environment associated with the paramagnetic surfaces
h_{jN}	Fraction of water relaxing in the j Nth pore environment associated with the nonparamagnetic surfaces
C_P	Mass fraction of iron-coated sand
C_N	Mass fraction of uncoated sand
ϕ	Porosity
m	Mass of the solid phase of the sample
r	Average radius of a sand grain
Relaxation Rate	
T_{2B}^{-1}	Bulk fluid relaxation rate
T_{2Sk}^{-1}	Surface relaxation rate for relaxation mode k
T_{2S0}^{-1}	The zeroth or slowest relaxation mode
$T_{2S\infty}^{-1}$	The fastest relaxation mode
T_{2j}^{-1}	Relaxation rate for pore environment j
$(T_{2S})_j^{-1}$	Surface relaxation rate for pore environment j
T_{2ML}^{-1}	Mean log relaxation rate
T_{2S}^{-1}	Average surface relaxation rate
T_{2jP}^{-1}	Relaxation rate for pore environment i for pores with paramagnetic content
T_{2jN}^{-1}	Relaxation rate for pore environment i for pores with no paramagnetic content
T_{2ML-P}^{-1}	Mean log relaxation rate for uniform geologic material composed of paramagnetic surfaces

Summary of mathematical symbols. (continued)

Symbol	Definition
T_{2ML-N}^{-1}	Mean log relaxation for uniform geologic material composed of nonparamagnetic surfaces
T_{2D}^{-1}	Diffusion relaxation rate
Surface Relaxivity	
ρ_2	Average surface relaxivity
ρ_{2P}	Surface relaxivity for the paramagnetic surface (i.e., iron-coated quartz sand surface)
ρ_{2N}	Surface relaxivity for the nonparamagnetic surface (i.e., quartz sand surface)
ρ_{2P-WC}	Surface relaxivity of the paramagnetic surface calculated assuming weak coupling
ρ_{2N-WC}	Surface relaxivity of the nonparamagnetic surface calculated assuming weak coupling
ρ_{2P-SC}	Surface relaxivity of the paramagnetic surface calculated assuming strong coupling
ρ_{2N-SC}	Surface relaxivity of the nonparamagnetic surface calculated assuming strong coupling.
ρ_{2F}	Surface relaxivity of the ferrihydrite-coated sand
ρ_{2H}	Surface relaxivity of the hematite-coated sand
Surface Area, specific surface area, surface-area-to-volume ratio, and volume	
S_{pore}	Surface area of a pore
S_N	Surface area of the nonparamagnetic surfaces
S_P	Surface area of the paramagnetic surfaces
S_{SN}	Specific surface area for the pure quartz sand
S_{SP}	Specific surface area for the iron-coated sand
S/V	Pore surface area to volume ratio for the measured sample
S_N/V	Ratio of the surface area of the nonparamagnetic surface to pore volume
S_P/V	Ratio of the surface area of the paramagnetic surface to pore volume
V_{pore}	Volume of a pore
V	Total pore volume in the measured sample
V_N	Volume of water within the pore space affected by the nonparamagnetic surfaces
V_P	Volume of water within the pore space affected by the paramagnetic surfaces
Control Parameter	
κ	Control parameter used to determine diffusion limits from average surface relaxivity
κ_{P-WC}	Control parameter calculated with the surface relaxivity value of the paramagnetic surface assuming weak coupling.
κ_{P-SC}	Control parameter calculated with the surface relaxivity value of the paramagnetic surface assuming strong coupling.

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